

(Universidade de São Paulo)
Pêndulo Simples e Pêndulo Duplo

Métodos Numéricos em Equações Diferenciais

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1 Pêndulo Simples

1.1 EDO

$$\mathbf{u}(t) = \begin{pmatrix} u_1(t) \\ u_2(t) \end{pmatrix} = \begin{pmatrix} q(t) \\ q'(t) \end{pmatrix} = \begin{pmatrix} q(t) \\ p(t) \end{pmatrix}, \quad q''(t) + \sin(q(t)) = 0 \iff p'(t) = -\sin(q(t))$$

$$\mathbf{u}'(t) = \begin{pmatrix} q'(t) \\ p'(t) \end{pmatrix} = \begin{pmatrix} p(t) \\ -\sin(q(t)) \end{pmatrix} = \begin{pmatrix} u_2(t) \\ -\sin(u_1(t)) \end{pmatrix} = \begin{pmatrix} f_1(\mathbf{u}(t), t) \\ f_2(\mathbf{u}(t), t) \end{pmatrix} = \mathbf{f}(\mathbf{u}(t), t)$$

$$\mathbf{u}^0 = \mathbf{u}(0) = \begin{pmatrix} q(0) \\ p(0) \end{pmatrix} = \begin{pmatrix} \frac{\pi}{4} \\ 0 \end{pmatrix}$$

1.2 Métodos

Para $h > 0$, $t_n = hn$, $\mathbf{U}^n = \begin{pmatrix} U_1^n \\ U_2^n \end{pmatrix} \approx \mathbf{u}(t_n)$ e $\mathbf{f}^n = \mathbf{f}(\mathbf{U}^n, t_n)$ temos:

1. Euler explícito: $\mathbf{U}^{n+1} = \mathbf{U}^n + h\mathbf{f}^n$;

$$2. \text{ RK4: } \begin{cases} K_1 = \mathbf{f}(\mathbf{U}^n, t_n) \\ K_2 = \mathbf{f}(\mathbf{U}^n + \frac{h}{2}K_1, t_n + \frac{h}{2}) \\ K_3 = \mathbf{f}(\mathbf{U}^n + \frac{h}{2}K_2, t_n + \frac{h}{2}) \\ K_4 = \mathbf{f}(\mathbf{U}^n + hK_3, t_n + h) \\ \mathbf{U}^{n+1} = \mathbf{U}^n + \frac{h}{6}(K_1 + 2K_2 + 2K_3 + K_4) \end{cases};$$

3. Euler implícito: $\mathbf{U}^{n+1} = \mathbf{U}^n + h\mathbf{f}^{n+1}$

1.2.1 Euler implícito

$$\mathbf{U}^{n+1} = \mathbf{U}^n + h\mathbf{f}^{n+1} \iff \mathbf{F}(\mathbf{U}) = \mathbf{U} - \mathbf{U}^n - h\mathbf{f}(\mathbf{U}, t_n) = 0$$

$$\mathbf{F}'(\mathbf{U}) = \frac{d\mathbf{F}}{d\mathbf{U}}(\mathbf{U}) = I_n - h\frac{d\mathbf{f}}{d\mathbf{U}}(\mathbf{U}, t_n)$$

onde I_n é a matriz identidade $n \times n$ e

$$\frac{d\mathbf{f}}{d\mathbf{U}}(\mathbf{U}, t_n) = \left(\begin{array}{cc} \frac{\partial f_1}{\partial U_1} & \frac{\partial f_1}{\partial U_2} \\ \frac{\partial f_2}{\partial U_1} & \frac{\partial f_2}{\partial U_2} \end{array} \right) \Big|_{(\mathbf{U}, t_n)} = \left(\begin{array}{cc} 0 & 1 \\ -\cos(U_1) & 0 \end{array} \right)$$

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$$\mathbf{F}'(\mathbf{U}) = \left(\begin{array}{cc} 1 & -h \\ h \cos(U_1) & 1 \end{array} \right)$$

Assim, com $\mathbf{U} = \mathbf{U}^{n+1}$ e um chute inicial $\mathbf{U}^{(0)}$, temos o método de Newton, para $k = 0, 1, \dots$ e com critério de parada $\|\delta_k\|_2 < 10^{-7}$:

$$\begin{cases} \mathbf{F}'(\mathbf{U}^{(k)})\delta_k = -\mathbf{F}(\mathbf{U}^{(k)}) \\ \mathbf{U}^{(k+1)} = \mathbf{U}^{(k)} + \delta_k \end{cases}$$

1.3 Gráficos

1.3.1 Posição angular pelo Tempo

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1.3.2 Ordem de convergência

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1.3.3 Velocidade angular pela Posição angular

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2 Pêndulo Duplo

2.1 EDO

$$\mathbf{u}(t) = \begin{pmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \\ u_4(t) \end{pmatrix} = \begin{pmatrix} q_1(t) \\ q_2(t) \\ p_1(t) \\ p_2(t) \end{pmatrix}$$

$$\mathbf{u}'(t) = \begin{pmatrix} q_1'(t) \\ q_2'(t) \\ p_1'(t) \\ p_2'(t) \end{pmatrix} = \begin{pmatrix} \frac{p_1 - p_2 \cos(q_1 - q_2)}{2 - \cos^2(q_1 - q_2)} \\ \frac{2p_2 - p_1 \cos(q_1 - q_2)}{2 - \cos^2(q_1 - q_2)} \\ -A_1 + A_2 - 2 \sin(q_1) \\ A_1 - A_2 - \sin(q_2) \end{pmatrix} \Big|_{(t)} = \begin{pmatrix} \frac{u_3 - u_4 \cos(u_1 - u_2)}{2 - \cos^2(u_1 - u_2)} \\ \frac{2u_4 - u_3 \cos(u_1 - u_2)}{2 - \cos^2(u_1 - u_2)} \\ -A_1 + A_2 - 2 \sin(u_1) \\ A_1 - A_2 - \sin(u_2) \end{pmatrix} \Big|_{(t)} = \mathbf{f}(\mathbf{u}(t), t)$$

onde

$$\begin{aligned} A_1(t) &= \frac{p_1 p_2 \sin(q_1 - q_2)}{1 + \sin^2(q_1 - q_2)} \Big|_{(t)} = \frac{u_3 u_4 \sin(u_1 - u_2)}{1 + \sin^2(u_1 - u_2)} \Big|_{(t)} \\ A_2(t) &= \frac{[p_1^2 + 2p_2^2 - 2p_1 p_2 \cos(q_1 - q_2)] \sin(q_1 - q_2) \cos(q_1 - q_2)}{[1 + \sin^2(q_1 - q_2)]^2} \Big|_{(t)} \\ &= \frac{[u_3^2 + 2u_4^2 - 2u_3 u_4 \cos(u_1 - u_2)] \sin(u_1 - u_2) \cos(u_1 - u_2)}{[1 + \sin^2(u_1 - u_2)]^2} \Big|_{(t)} \end{aligned}$$

2.2 Trajetória de um pêndulo

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2.3 Experimentos

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